

Predicting Food Security Index & PROMOTE SUSTAINABLE AGRICULTURE

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Abstract

Food insecurity remains a persistent challenge, even in high-producing agricultural systems. Timely national-level indicators are needed to support monitoring and policy decisions. This study tests whether U.S. agricultural production signals can predict a composite Food Security Index (*FSI*) derived from food-system outcomes. We constructed an annual United States dataset spanning from 1970 to 2023 that integrates USDA NASS QuickStats measures for three major staple crops, corn, soybeans, and wheat (yield, production, area planted, and area harvested).

The prediction target is an *FSI* scaled from 0 to 1, where higher values indicate lower food insecurity, constructed through mathematical modeling of four FAOSTAT variables: dietary energy supply, protein supply, GDP per capita, and a food production index. We compared multiple supervised regression models using a fixed split of 1970 to 2012 data for training, and a testing period of 2013 to 2023. We further included a change-based formulation that predicts a year to year change to *FSI* levels (ΔFSI), with engineered shock and lag features.

A Random Forest model trained on the ΔFSI feature set performed the best, achieving a Root Mean Squared Error (RMSE) of about 0.066 on the 2013-2023 testing data and improving on a persistence baseline. Overall, results showed that the national crop statistics contain measurable predictive signals for annual food security dynamics and motive extensions.

Keywords:

Agriculture, Food Security Index, Prospective Validation

1 Introduction

[1, 2, 3], [4], [5, 6, 7]
[8], [6], [9], [7], [10]
[11]

Even in highly productive agricultural systems, food insecurity is still a continuous challenge as national food security depends on more than the total output of crops. Beyond aggregated supply, food security is a reflection on whether or not people can consistently access safe, sufficient, and nutritious food while the system remains stable under shocks. As climate variability, market disruptions, and economic pressures interact, there is an increased interest in timely, interpretable national indicators that can be monitored over long periods in support of decision making.

Agricultural production is one of the most consistently measured national variables. In the United States, corn, soybeans, and wheat are especially well documented as they dominate staple crop production and influence downstream food and feed markets. Yearly changes in yield, acreage, and total production have the potential to capture biophysical constraints and economic land-use responses. However, directly linking production to national food security outcomes is not straightforward; food security also depends on economic access, supply chains, and broader macro conditions, which introduce nonlinear relationships and time-lagged effects.

2 Materials & Methods

2.1 Data and Study Design

We evaluated whether or not national staple crop production can predict a composite Food Security Index (*FSI*) for the United States

(U.S.) using an annual time series (1970-2023). Three different crop predictors were compiled from the United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) Quickstats: corn, soybeans, and wheat, including yield, production, and harvested area for each crop ($N=1,944$). The prediction target was an *FSI* scaled to $[0, 1]$, where a higher value signaled better food security, computed from four different components derived from the Food and Agriculture Organization Corporate Statistical (FAOSTAT) Database: dietary energy supply (kcal/capita/day), protein supply (g/capita/day), GDP per capita, and a food production index.

In addition to crop exclusive statistics, we included a set of exogenous drivers to capture broader conditions that can influence food security and crop performance. U.S. food-price conditions were represented using a food Consumer Price Index (CPI) index and its derived annual change. The World Bank Commodity Price Data (“Pink Sheet”) indices (Energy, Agriculture, Food, Grains, Fertilizers, Metals & Minerals) and selected commodity price series (crude oil, maize, soybeans, and U.S. wheat prices: Hard Red Winter, HRW, and Soft Red Winter, SRW) acted as global input and commodity pressures.

All raw datasets were first restricted to the United States and then converted to annual values and merged by year to form a single master table. Non-numeric values were treated as missing and excluded only when required for the given model fit to maintain a consistent predictor-target alignment. The resulting dataset was then used for feature engineering (lags and year-over-year “shock” variables) and model evaluation under time-respecting train/test splits.

2.2 Targets and Feature Engineering

We tested two prediction formulations: level and change-based. In the level formulation, models predict $FSI(t)$ directly, where $FSI(t)$ denotes the *FSI* in year t . In the change based formulation, models predict annual change $\Delta FSI(t) = FSI(t) - FSI(t-1)$ and reconstruct levels via $FSI(t) = FSI(t-1) + \Delta FSI(t)$. Where this year’s predicted *FSI* equals last year’s *FSI* plus the model’s predicted change this year. Predictors included crop indicators (yield, production, and area harvested for corn, soybeans, and wheat) as well as exogenous variables in a climate and economic context.

To capture disruptions and delayed effects,

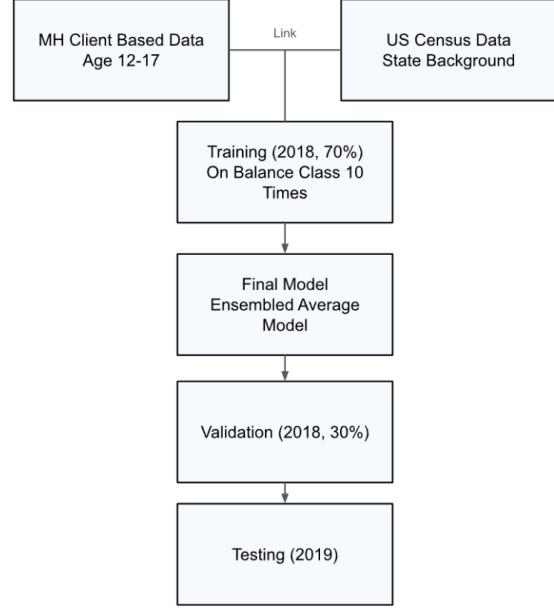


Figure 1: Study Flow Chart

we manually created shock features and lags for all time-varying predictors. Shocks were computed on a year by year basis as changes .(percentage changes for strictly positive variables) We also included 1 to 2 year lags of both raw and shocked level terms. This design allows our model to learn immediate versus delayed associations between production, external conditions, and changes in national food security.

2.3 Performance Metrics

Predictive accuracy on held-out years was measured using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 values, reported in relevance to the persistence baseline. For the second formulation, the change-based formulation, we assessed directional performance additionally by classifying whether ΔFSI was bigger than 0 and then computing a confusion matrix and balanced accuracy along with it. Receiver Operating Characteristic Area Under the Curve (ROC AUC) and average precision were also calculated using the changed, predicted scores.

Figure 1 presents the flow chart for the machine learning model, including the data integration and model validation.

Notably, the MH-CLD data does not link individual case IDs across years, ensuring that cases are not traceable longitudinally. To minimize bias and maintain model integrity, no OUD data were included in the features. This approach ensured a clean separation of training, validation, and testing data to support robust model devel-

opment and evaluation.

2.4 Data Preparation

The data preparation process was conducted using Python 3.11 [12] with the scikit-learn library [13] for predictive modeling. The primary mental health features included diagnoses such as ADHD, anxiety, depressive disorder, oppositional defiant disorder (ODD), pervasive developmental disorder (PDD), conduct disorder, bipolar disorder, schizophrenia, and other mental health conditions. All variables, except for the count of mental health diagnoses, were numerical.

Demographic features including age group, sex, state, race, and ethnicity, were categorical and were one-hot encoded into binary variables. Substance use diagnosis variables provided outcome information and included major conditions such as opioid, alcohol, cannabis, cocaine, and other substance-related disorders. Figure 2 summarizes the descriptive information of the 2018 cohort. Background features such as population size, economy, education, income information, and so on were derived from external datasets (detailed in Supplementary Table A1). To ensure consistent scaling across variables with different units, all continuous features were normalized using the StandardScaler from scikit-learn’s preprocessing package. This normalization improved the comparability of features and optimized model performance.

2.5 Model Selection and Evaluation

The COVID-19 pandemic has led to varying social and environmental conditions across different periods, particularly from 2020 to 2022. These changes can influence model performance, making it challenging to generalize findings from pre-pandemic data, such as 2018, to the pandemic era. To address this, we first trained the model on the 2018 data and tested it on the 2019 dataset. This approach helps evaluate the model’s performance in a pre-pandemic context. Additionally, we trained and tested the model exclusively using the 2022 data to account for the relatively stable social factors specific to that period during the pandemic. Evaluating the model on both pre-pandemic (2018 and 2019) and post-pandemic (2022) data aimed to determine whether the key risk factors contributing to OUD were consistent across different personal, social, and environmental contexts.

For the 2018 data and 70% of 2022 data (30% holding for testing), the dataset was split into

a 70%/30% ratio using stratified sampling to address the class imbalance, ensuring that the proportion of subjects with opioid use disorder was consistent in both the training and validation sets. A logistic regression and Lasso model from the scikit-learn library were employed with balanced class weights configured to account for the unequal distribution of outcomes. To ensure robust performance evaluation, we repeated the training and validation process 10 times. The final model is built based on an ensemble model by averaging out each iteration’s coefficients. Performance metrics, including sensitivity, specificity, balanced accuracy, and the area under the receiver operating characteristic curve (ROC-AUC) were used to provide a comprehensive assessment of the model’s performance. Feature importance was also calculated to identify key predictors. The model was then tested with the 2019 data and the rest of the 2022 data. This allowed us to assess the machine learning algorithms’ generalization in the pre- and post-pandemic period.

2.6 Assessing Feature Importance

The importance of features for predicting OUD was evaluated based on the model coefficients. Features with larger absolute coefficient values were deemed more significant, indicating a stronger influence on the target outcome. The sign of the coefficients further reflected the direction of the association: a positive coefficient signified a positive association with OUD, whereas a negative sign indicated an inverse relationship.

3 Results

3.1 Visualizing Trends of Youth Mental Health Prevalence

Figure 3 illustrates the trends in mental health diagnoses over the past decade, spanning from 2013 to 2022. The analysis focuses exclusively on conditions demonstrating an upward trend, including substance use problems (??), anxiety disorders (??), depressive disorders (??), and trauma-related (??) conditions. Other mental health conditions were presenting either a downward trend or insufficient trending, rendering them less significant for this analysis.

Figure 4 presents geospatial heat maps from QGIS [14] illustrating life expectancy (??), rent costs (??), actual OUD cases in 2019 (??), and predicted cases using Logistic Regression (??). The heat maps for life expectancy and rent costs reveal clusters of hot and cold spots predominantly along coastal regions, particularly in the

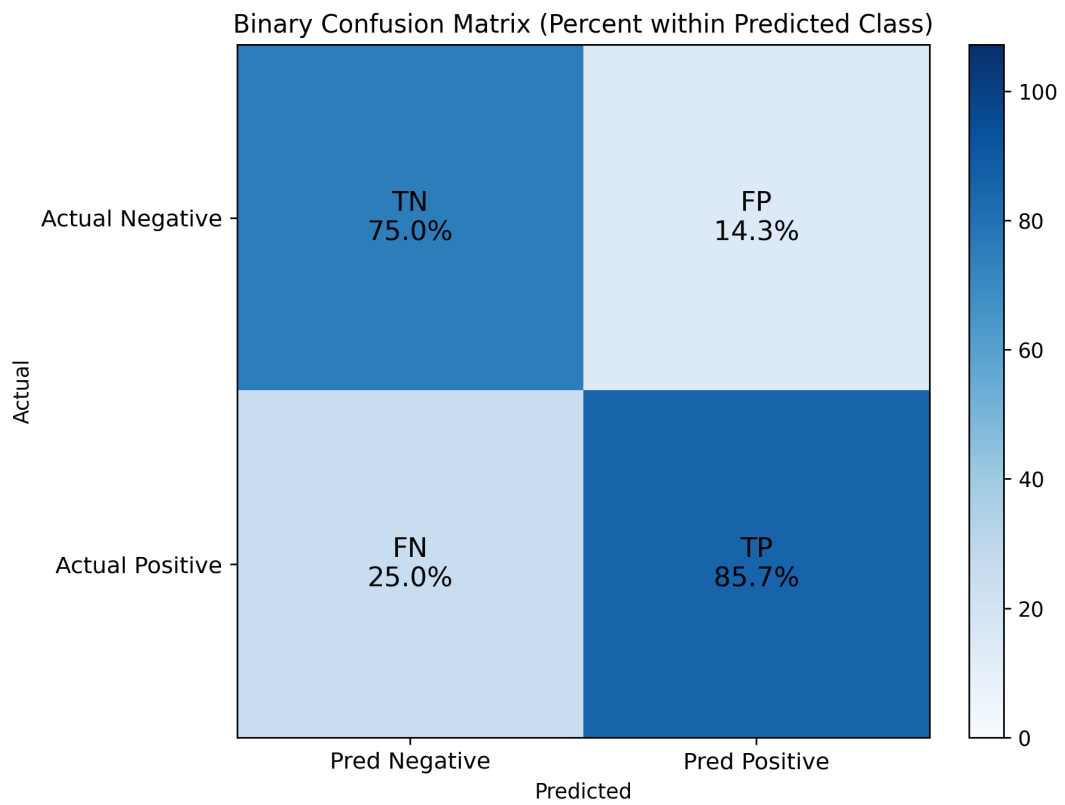


Figure 2: Confusion Matrix

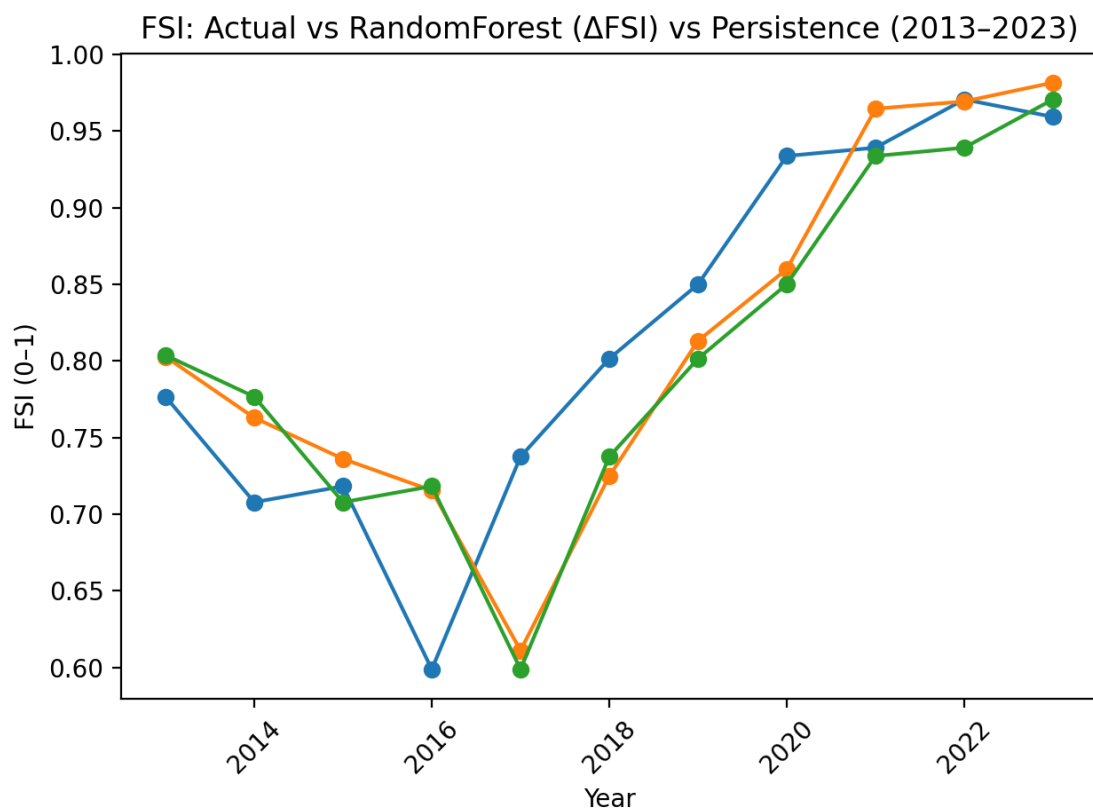


Figure 3: Comparison of different prediction methods

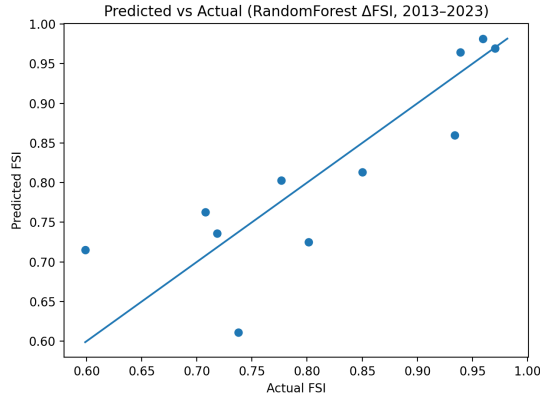


Figure 4: Scatter plots between Actual and Predicted data for 2013-2023

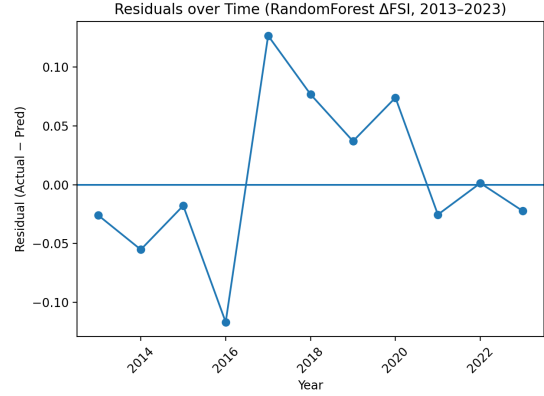


Figure 5: Residual plot over time period 2013-2023

western US and New England. These patterns align with certain hotspots observed in the heat map of actual OUD cases. Moreover, the heat map of predicted cases highlights the improved performance of the machine-learning model.

3.2 Model Performance

OUD was most effectively predicted using a logistic regression classification algorithm (Ridge classifier) compared with the LASSO, a regression technique that performs both feature selection and regularization by shrinking less important feature coefficients to zero, helping simplify models and preventing overfitting, classifier, and Random Forest classifier (See Figure 5). Logistic regression achieved a balanced accuracy of 76.6%, compared to 76.5% for LASSO and 61.0% for Random Forest.

On the 2018 validation data, the ensemble logistic model achieved a balanced accuracy of 73%, with a sensitivity of 75% and specificity of 71% with an area under the curve (AUC) of 0.81. For the 2019 testing data, the model demonstrated a balanced accuracy of 50%, with a sensitivity of 16% and specificity of 84%. On the 2022 testing data, the model achieved a balanced accuracy of 86%, sensitivity of 84%, and specificity of 87%. The model has an area under the curve (AUC) of 0.82. The ROC curves (??, ??) and confusion matrix (??, ??) plots for both the 2019 testing data and 2022 validation data are presented in Figures 6 and 7, respectively.

3.3 Feature Importance

Figure 8 presents the top 15 dominant features in the 2018 and 2022 datasets using the logistic regression classifier. The absolute values of the coefficients indicate feature importance with higher absolute values signifying greater impor-

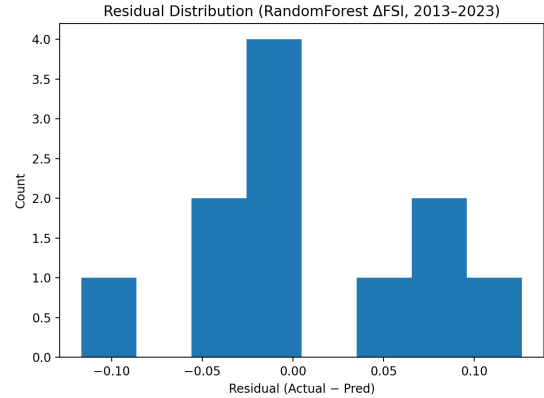


fig6_roc_bootstrap10

Figure 6: Histogram of Residual Errors

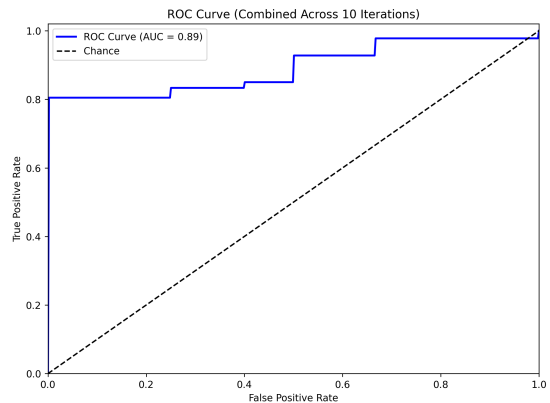


Figure 7: ROC curve for 2022 Training model

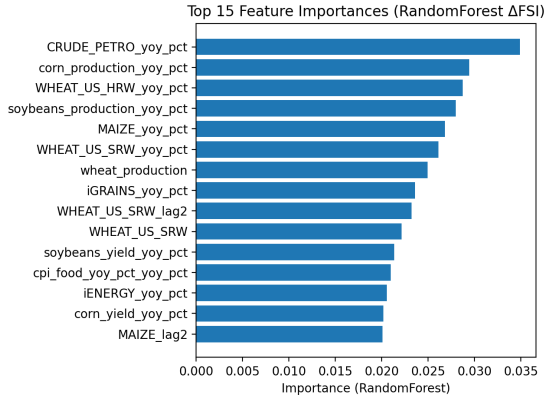


Figure 8: Top 15 most important features for 2018 and 2022 testing data using Logistic Regression

tance. The sign of the coefficients reflects the direction of the association with OUD; positive values indicate increased risk while negative values suggest protective effects. For instance, living in the District of Columbia, Puerto Rico, or North Carolina is associated with lower OUD incidence whereas residing in Virginia, Maryland, or Minnesota corresponds to a higher risk.

Two social determinants positively associated with OUD were identified: life expectancy and the percentage of children in the state. Notably, there were no significant changes in the coefficients or feature rankings between 2018 and 2022. The complete list of all features with ranking can be found in Supplementary Table A1.

4 Discussion

Our analysis yielded the logistic regression classifier with ridge regularization as the best in predicting OUD. This classifier demonstrated strong performance in identifying OUD cases in 2018 and 2022, achieving satisfactory specificity even in 2019 despite slightly lower other results. Consistent geographic factors were identified as top predictors across 2018 and 2022, underscoring the significant impact of local geographic variations on the risk of SUD.

The use of the logistic regression classifier (Ridge) is based entirely on a data-driven approach. To ensure the selected model accommodates diverse data types, the model selection process included both tree-based Random Forest and non-tree models like Lasso and Ridge logistic regression classifiers. This approach minimizes the risk of arbitrary model selection due to unknown relationships between predictors and the target variable. Given the highly imbalanced nature of the dataset, class-weight adjust-

ments were applied to account for the minority class, reducing potential classification bias. These measures enhance the robustness and objectivity of the study results.

Our findings indicate sufficient predictive performance within the same year, such as in 2018 or 2022, but poor performance across different years, such as using 2018 data to predict 2019 outcomes. This discrepancy may stem from significant variations in opioid-related variables across different states and years. For instance, data from the CDC in the United States demonstrate that the opioid dispensing rate in the United States varies across states and over time. Such changes likely reflect distributional discrepancies (data drift) between the OUD population in the 2018 training set and the 2019 test set, reducing cross-year predictive performance. This effect is more pronounced in the current imbalanced datasets with limited target samples, where minor variations in sample distribution across years can significantly impact model performance. To address this limitation in future studies with larger datasets, incorporating temporal features (e.g., rates of change) and developing year-based stratified models could more effectively capture cross-temporal variations.

Geographic data thereby emerged as an important predictor of variation in the diagnosis of OUD amongst adolescents receiving mental health services and reflected state-by-state differences in policy, economy, and environment. Aggregated patterns at the state level showed similarities in the course of illness while the integration of the U.S. Census provided important contextual information on social determinants and socioeconomic variables explaining the risk of developing OUD. The absence of critical variables such as public health policy changes, intervention efforts, and governance, limited our findings and should be addressed in future research.

Therefore, this model is specific to youth who have accessed mental health services and thus may not be generalized to the broader adolescent population. Future iterations may wish to extend this methodology to include broader population health information. We excluded the substance use diagnosis variables like alcohol, cocaine, or cannabis use in order to minimize bias as these variables are not independent in defining OUD within this dataset.

Even the simplified model, using only individual mental health diagnoses and service utilization variables, achieved highly balanced accuracy without geographic or social determinant data. Apart from geographic factors, mental health conditions, such as, anxiety, depres-

sion, and trauma emerged as strong predictors of OUD in adolescents due to the interconnected nature of mental health conditions and substance use. Adolescents with mental health challenges may experience emotional dysregulation, heightened stress, and a lack of coping mechanisms, leading to self-medication behaviors. Opioids, often perceived as quick relief from psychological pain, are more likely to be misused in this vulnerable population. Additionally, trauma, such as adverse childhood experiences (ACEs), is strongly linked to increased susceptibility to substance use as a coping strategy, further amplifying the risk of OUD.

Adolescents benefit from preventive strategies focused on resilience-building and coping skills. Tailoring interventions to the developmental context and life circumstances of each population is crucial for effective prevention and treatment. Our study provides individualized predictions that can be aggregated at various levels to inform targeted, cost-effective prevention programs. This approach enables U.S. local governments to better address the complex interplay between local conditions and OUD risk.

5 Conclusion

Youth receiving mental health services face heightened risks of developing OUD due to complex social, psychological, and environmental challenges. Early detection and intervention are crucial for reducing their vulnerability to substance use disorders. Our model shows great potential to identify the risk of OUD in this group and offer timely support to prevent further advancement of substance use disorders.

Excluding prediction power, our analysis reveals significant geographic variation in OUD prevalence, driven by state policies, economic conditions, and healthcare access. This highlights the need for geographically tailored interventions to ensure equitable resource distribution and effective utilization of resources. Enhancing predictive tools from this study can offer insights to guide government policies in reducing such disparities.

Through embedding advanced analytic into policy making, governments can more effectively identify at-risk populations and design community-level strategies that address the root causes of OUD. This approach holds the promise of addressing inequities in the opioid crisis while empowering healthcare systems to take a more proactive stance. It paves the way for more effective and equitable outcomes for youth navigating mental health challenges and substance use disorders, fostering a fairer and more responsive

support network.

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6 Appendix A

6.1 Supplementary Table A1: All Features and Ranks

Rank	Features	Description
1	STATEFIP_11	District of Columbia
2	STATEFIP_51	Virginia

Table A1: Feature Detail List